1. **Cover Page**
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HUMAN COMPUTER INTERACTION SECURITY (HCI711S)	
Lab :	Individual Assignment
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Due Date:	10 April 2026

Title

An Analysis of User Interface Design, User Experience, and Security Conflicts in Modern Computer Interfaces

1, Introduction

Human Computer Interaction (HCI) security focuses on how users interact with computer systems while maintaining usability and protection of information. In modern computing, interfaces must be easy to use, efficient, and secure at the same time. However, there are often conflicts between user convenience and security requirements. This paper analyses the IBM/Lenovo ThinkPad TrackPoint and Trackpad as the main example, together with two additional interfaces: the smartphone fingerprint scanner and the ATM touch-screen interface. The discussion highlights good and bad aspects of each interface, evaluates the conflicts between UX and security, and recommends improvements.

User interface design principles such as consistency, visibility, feedback, simplicity, and accessibility play an important role in how users experience a system. At the same time, security requirements such as authentication, confidentiality, and access control must be integrated without negatively affecting usability.

2. Background and Literature Review

Before conducting this research, it was already known that interface design directly affects both productivity and security. According to Norman (2013), good design should make systems intuitive and reduce the chances of user error. Similarly, Schneier (2015) explains that many security failures are caused by poor interaction design rather than weak technical controls.

In HCI security, the main challenge is balancing ease of use with strong protection mechanisms. For example, an interface that requires too many authentication steps may frustrate users, while a very simple system may expose sensitive information.

3. Interface Analysis

3.1 IBM/Lenovo ThinkPad TrackPoint and Trackpad

Image of thinkpad trackpoint and trackpad:

Good Aspects (UI and UX)

The ThinkPad TrackPoint is a small pointing stick located between the keyboard keys. One major advantage is efficiency. Users can move the cursor without removing their hands from the keyboard, which improves speed and productivity.

The trackpad also provides a familiar and intuitive interaction method. It supports gestures such as scrolling, zooming, and multi-touch commands, which improve user experience.

Other positive UI/UX principles include:

- Efficiency: quick cursor control while typing.
- Consistency: common laptop navigation design.
- Feedback: visible cursor movement based on user input.
- Accessibility: supports multiple input preferences.

Bad Aspects (UI and UX)

The TrackPoint may be difficult for new users to understand because it is smaller and less obvious than a mouse or trackpad.

The trackpad may also cause accidental touches while typing, which can move the cursor unexpectedly and disrupt workflow.

Weaknesses include:

- Steep learning curve for beginners.
- Small size may reduce precision.
- Accidental cursor movement.
- May not be comfortable for prolonged use.

3.2 Other Interface 1: Smartphone Fingerprint Scanner

Image of smartphone fingerprint scanner :

Good Aspects

The fingerprint scanner provides quick and secure access to a device. It improves convenience because users do not need to remember passwords.

UI/UX strengths:

- Fast authentication.
- Easy to use.
- Immediate visual feedback.
- Enhances user trust.

Bad Aspects

Sometimes the scanner fails to recognize the fingerprint due to moisture, dirt, or poor finger placement.

Weaknesses:

- Recognition errors.
- Can frustrate users.
- Accessibility issues for injured fingers.

3.3 Other Interface 2: ATM Touch-Screen Interface

Image of atm touch-screen interface:

Good Aspects

ATM touch screens are designed to be simple and task-oriented. Large buttons and guided steps make the system easy for most users.

UI/UX strengths:

- Clear navigation.
- Large readable options.
- Guided workflow.
- Suitable for public use.

Bad Aspects

Poor screen responsiveness or confusing menu structures can delay transactions.

Weaknesses:

- Screen glare in sunlight.
- Slow response times.
- Limited accessibility for visually impaired users.

4. UX and Security Requirement Conflicts

4.1 ThinkPad TrackPoint and Trackpad

The major conflict is between convenience and accidental actions. While the interface improves efficiency, accidental cursor movement can lead to unintended clicks, file deletion, or sending incorrect information.

From a security perspective, accidental clicks may cause users to open malicious links or approve unwanted actions.

4.2 Fingerprint Scanner

The conflict here is between security strength and usability. Strong biometric authentication improves security, but recognition failures reduce user satisfaction.

If the system is too strict, legitimate users may be denied access.

4.3 ATM Touch Screen

The conflict is between privacy and usability. Large clear screens improve usability, but they may expose sensitive information such as account balances or PIN prompts to nearby people.

This creates risks such as shoulder surfing.

5. Suggested Improvements

5.1 ThinkPad TrackPoint and Trackpad

Suggested improvements include:

- Increase TrackPoint sensitivity customization.
- Add palm rejection for the trackpad.
- Provide first-time tutorial prompts.
- Improve tactile feedback.

5.2 Fingerprint Scanner

Suggested improvements:

- Add fallback PIN or facial recognition.
- Improve sensor accuracy.
- Provide clearer retry feedback messages.

5.3 ATM Touch Screen

Suggested improvements:

- Add privacy screen filters.
- Improve screen brightness adjustment.
- Use voice-guided accessibility options.
- Simplify menu navigation.

6. Conclusion

This paper has shown that good interface design must balance usability and security. The ThinkPad TrackPoint and trackpad improve efficiency but may introduce accidental input risks. The fingerprint scanner improves authentication convenience but may frustrate users when recognition fails. The ATM interface is simple and effective but may expose sensitive information.

The major lesson learned is that secure systems must remain user-friendly. If security controls are too difficult to use, users may avoid them or make mistakes. Future interface designs should focus on accessibility, usability, and stronger protection mechanisms.

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